

Operating systems

Commands

Content Lecture#7

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Lets take a Break

UNIX is user friendly...

It's just very particular about **who its friends are**.

Why computers are like human:



1. In order to get their attention, you have to turn them on.
2. They have a lot of data, but are still clueless.
3. They are supposed to help you solve problems, but half the time they are the problem.
4. As soon as you commit to one, you realize that if you had waited a little longer, you could have had a better model.

Real programmers count from **0**.

What code does a depressed programmer write?
"Goodbye world!"



Threads

Flow of Execution
Basic Unit of execution

POSIX

Postable Operating System interface for Unix (POSIX),

Threads

- Thread idea is used for higher throughput and better performance
- Threads are cheaper in terms of system overheads
- Avoid context switching of processes if thread switch is used
- Each thread run sequentially
- Thread can create child threads
- If thread is blocked, another thread within same process can run
- A traditional or heavyweight process is equal to a task with one thread
- Threads provide a mechanism to achieve parallelism
- Threads are very important in the design of distributed and client server systems
- In some systems, threads are supported by the operating system
- Where the OS is not multithreaded, threads cannot be managed with application system code
- Some database management systems provide their own thread management
- Some languages, like Java, provide direct support for writing the thread based programs
- In the absence of language support, threads can still be utilized by use of library subroutines

Benefits of Multithreading

- **Responsiveness**
  - It allows to run a program even if some part of it is in block state
- **Resource sharing**
  - Different threads of a program behaving like multiple processes are sharing same address space
- **Economy**
  - Overheads of context switching of TCB are much lesser than of PCB
- **Scalability**
  - Utilization of multiprocessor architecture
    - Threads are very useful in multiprocessor architecture
  - Due to threading a single program can run on different processors at same time

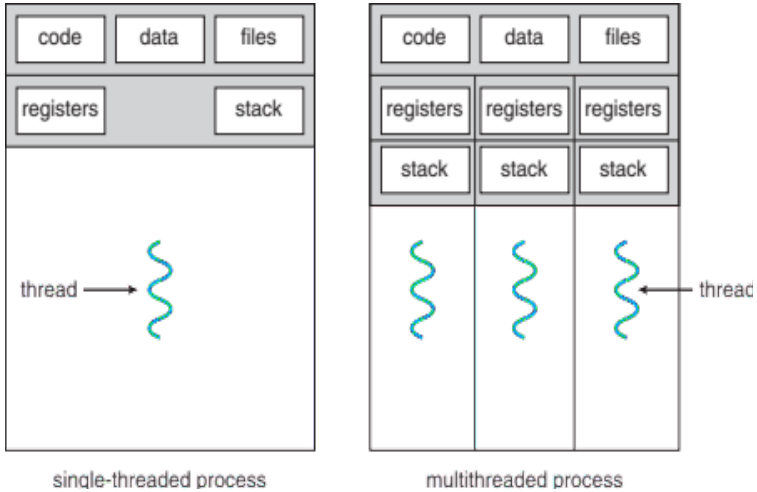
Why Threads

- After Fork(), a child process is created which is a copy of parent process.
- The child Process contains the same code as of parent process.

- Why to replicate Process?
- So that different parts of a program come to execution.
- Each process has 1 flow of execution.
- Can we achieve the same without replicating the address space?

Threads VS Process

- Multi threads shares code, data, heap and file descriptors are shared.
- Every thread has its own registers and stack
- Two threads can communicate without Pipe because data and segment is easy and can be done using global variable or using file descriptor one thread can read and other can write.





Thank you